

ASSIGNMENT CLA-01

Circuit Design (SEC301)

DC-DC Boost Converter Design And Analysis

Submitted by

Kamineni Dhana Sri Keerthi

AP24110040009

Department of Electrical and Electronics Engineering
SRM University AP

Submitted to

Dr Lohith Kumar Pittala

Assistant Professor

Department of Electrical and Electronics Engineering
SRM University AP



SRM University
Andhra Pradesh

Department of Electrical and Electronics Engineering
School of Engineering and Sciences (SEAS).

August 2026

TABLE OF CONTENT

S.NO.	CONTENT	PAGE NO.
1	Boost Converter Introduction	2
2	Circuit Diagram	2
3	Switching Operation	2-3
4	Waveforms of Boost Converter	4
5	Analytical Derivations (a) Average Output Voltage (CCM & DCM)	5,7
6	Analytical Derivations (b) Inductor current ripple (CCM & DCM)	6,9
7	Analytical Derivations (c) Capacitor Voltage ripple (CCM & DCM)	6,11
8	Analytical Derivations (d) Critical Resitance (CCM & DCM)	6
9	Matlab Simulations (a) CCM	11-14
10	Matlab Simulation (b) DCM	14-16
11	Matlab Simulation (c) Boundary Condition	18-19

0.1 Boost Converter Introduction

01. A Boost converter is a DC-DC converter that convert lower DC input to higher DC Output.

02. It is also known as a step up converter. The switch in the converter is operated at very high frequency using Pulse Width Modulation.

0.2 Boost Converter Circuit Diagram

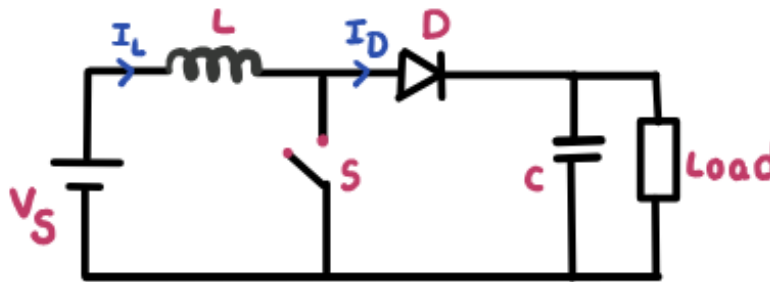


Figure 1: Circuit Diagram

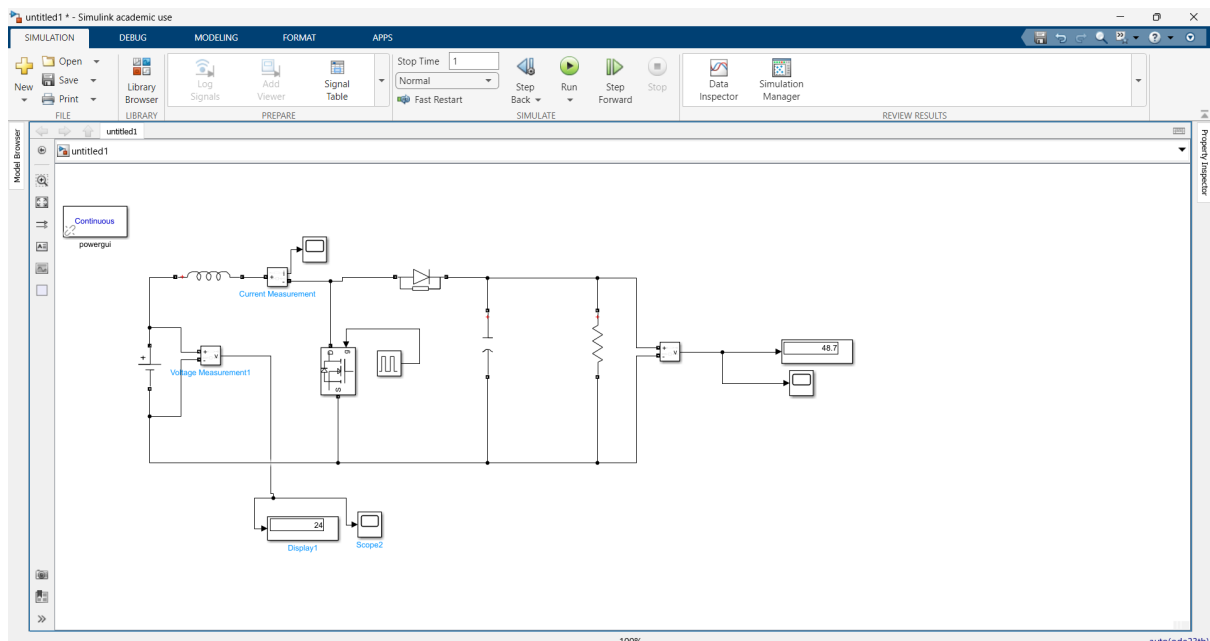


Figure 2: Matlab version circuit diagram

0.2.1 Switching Operation

The switching operation of a boost converter consists of two modes: **ON-state** and **OFF-state**.

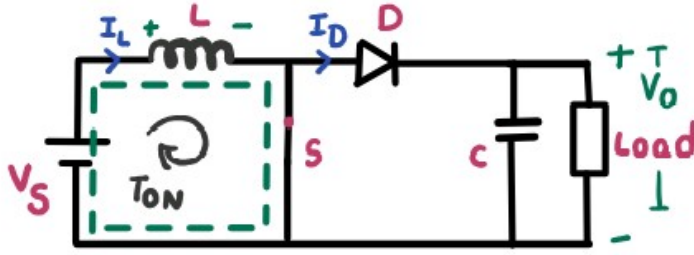


Figure 3: On state Current flow

Switch ON

When the switch S is ON, the current follows the path shown in Fig. 3. The diode D is reverse biased and therefore does not conduct. The input voltage is directly applied across the inductor, causing the inductor current to increase linearly.

$$V_L = V_{in} \quad (1)$$

Since

$$V_L = L \frac{di_L}{dt}, \quad (2)$$

The capacitor supplies the load during this interval.

Switch OFF

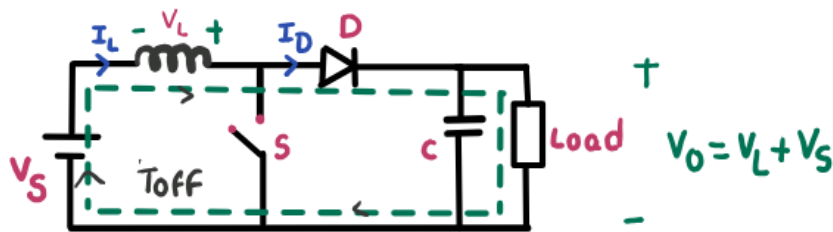


Figure 4: OFF State Current Flow

When the switch S is OFF, the current follows the path shown in Fig. 17. The diode D becomes forward biased and conducts. The energy stored in the inductor is transferred to the capacitor and load.

The inductor voltage during the OFF interval is

$$V_L = V_{in} - V_o. \quad (3)$$

Since $V_o > V_{in}$ for a boost converter,

$$V_L < 0, \quad (4)$$

0.3 Waveforms

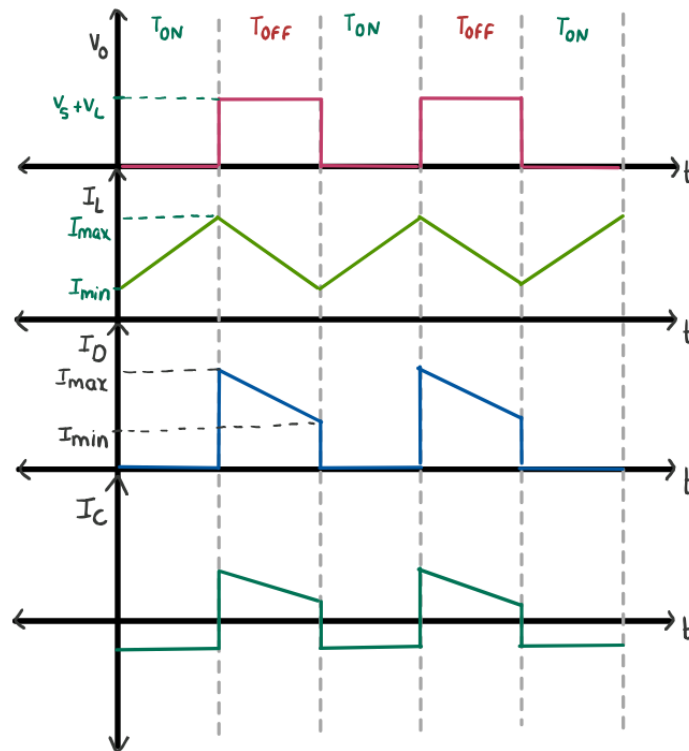


Figure 5: Waveforms

1. In V_o vs t graph during switch ON time, $V_o = 0$ and during OFF time,

$$V_o = V_s + V_L$$

2. In I_L vs t graph during switch ON time ($0 < t < DT$), and during OFF time ($DT < t < T$) decreases linearly.

1. During ON

$$V_L = V_{in} \quad \& \quad V_L = L \frac{di_L}{dt}$$

$$\Rightarrow \frac{V_{in}}{L} = \frac{di_L}{dt}$$

Since V_L & L are constants, the slope is constant.

2. During OFF

$$V_L = V_{in} - V_o \quad \& \quad V_L = L \frac{di_L}{dt}$$

$$\Rightarrow \frac{V_{in} - V_o}{L} = \frac{di_L}{dt}$$

Since $V_o > V_{in}$, the slope is negative.

0.4 Analytic Derivations

0.4.1 CCM Derivations

Average Output Voltage

Let us assume:

V_{in} = input voltage

V_o = average output voltage

L = inductor

D = duty ratio, where $D = \frac{T_{ON}}{T_s}$

T_s = switching period = $\frac{1}{f_s}$

Volt-second balance:

$$V_{in}(DT_s) + (V_{in} - V_o)(1 - D)T_s = 0$$

Dividing by T_s ,

$$V_{in}D + (V_{in} - V_o)(1 - D) = 0$$

$$V_{in}D + V_{in}(1 - D) - V_o(1 - D) = 0$$

$$V_{in} = V_o(1 - D)$$

Therefore,

$$V_o = \frac{V_{in}}{1 - D}$$

Inductor Current Ripple

$$V_L = V_{in} \quad V_L = L \frac{di_L}{dt} \quad \text{Switch ON}$$

$$\frac{V_{in}}{L} = \frac{di_L}{dt}$$

$$\Delta I_{L,ON} = \frac{V_{in}}{L} DT_s$$

$$\Delta I_L = \frac{V_{in}D}{Lf_s} \quad \text{peak to peak}$$

$$V_L = V_{in} - V_o \quad \text{Switch OFF}$$

$$\Delta I_{L,OFF} = \frac{V_{in} - V_o}{L}(1 - D)T_s$$

$$-\Delta I_L = \frac{V_{in} - V_o}{L}(1 - D)T_s$$

$$\Delta I_L = \frac{V_{in}D}{Lf_s} \quad \text{where } V_{in} = V_o(1 - D)$$

$$\Delta I_L = I_{L,max} - I_{L,min}$$

where

$$I_{L,max} = I_{L,avg} + \frac{\Delta I_L}{2}$$

$$I_{L,min} = I_{L,avg} - \frac{\Delta I_L}{2}$$

Capacitor Voltage Ripple

$$i_C = -I_o \quad \text{diode OFF, capacitor supplies load}$$

$$T_{ON} = DT_s = \frac{D}{f_s}$$

$$i_C = C \frac{dV_C}{dt}$$

$$\Delta V_C = \frac{I_o T_{ON}}{C}$$

$$\Delta V_o = \frac{I_o D}{C f_s}$$

$$\Delta V_o = \frac{V_o D}{RC f_s} \quad \text{for } I_o = \frac{V_o}{R}$$

Critical Resistance

$$I_{L,min} = 0 \quad \text{At the boundary between CCM \& DCM}$$

$$I_{L,avg} = \frac{I_{L,max} + I_{L,min}}{2}$$

$$I_{L,avg} = \frac{I_{L,max}}{2}$$

$$V_{in} I_{in} = V_o I_o$$

$$I_{in} = \frac{V_o I_o}{V_{in}}$$

$$I_{L,avg} = \frac{V_o I_o}{V_{in}}$$

$$I_{L,avg} = \frac{V_o^2}{V_{in} R} \quad \text{where } I_o = \frac{V_o}{R}$$

$$I_{L,avg} = \frac{V_{in}}{(1-D)^2 R}$$

$$\Delta I_L = \frac{V_{in} D}{L f_s}$$

$$\frac{\Delta I_L}{2} = I_{L,avg}$$

$$\frac{V_{in} D}{2 L f_s} = \frac{V_{in}}{(1-D)^2 R}$$

$$R_{critical} = \frac{2 L f_s (1-D)^2}{D}$$

$$R < R_{crit} \Rightarrow CCM$$

$$R > R_{crit} \Rightarrow DCM$$

0.4.2 DCM Derivations

Average Output Voltage

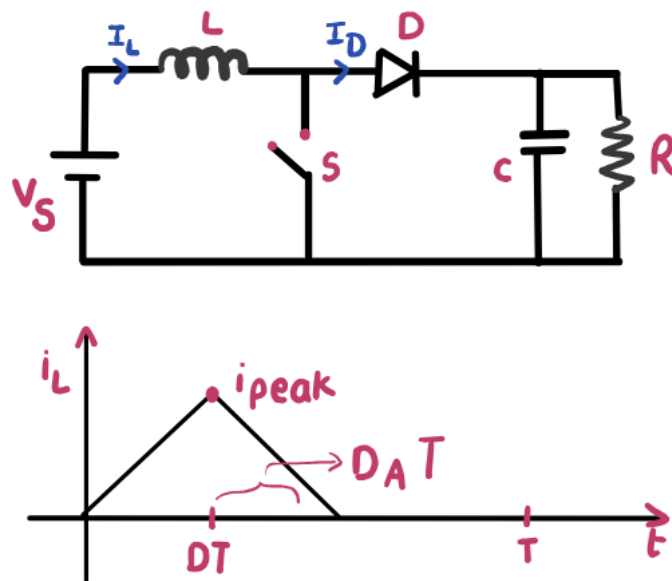


Figure 6: DCM Average Output Voltage graph

$V_{in}I_{in} = V_oI_o$ By power balance, assuming ideal condition

$$V_{in}I_{in} = \frac{V_o^2}{R}$$

$$I_o = \frac{V_o}{R}$$

According to the graph,

$$I_{L,avg} = \frac{1}{2}(D + D_A)I_{peak}$$

$$I_{L,avg} = \frac{1}{2}(D + D_A)I_{peak}$$

$$I_{L,avg} = \frac{V_o^2}{V_{in}R}$$

Now to find I_{peak} ,

$$V = L \frac{\Delta i}{\Delta t}$$

$$\Delta i = \frac{V\Delta t}{L}$$

From diagram,

$$I_{peak} = \frac{V_{in}DT}{L}$$

Now D_A ,

$$V = L \frac{\Delta i}{\Delta t}$$

$$V_{in} - V_o = -\frac{I_{peak}}{D_A T}$$

$$D_A = \frac{-I_{peak}}{T(V_{in} - V_o)}$$

$$D_A = \frac{I_{peak}}{T(V_o - V_{in})}$$

$$D_A = \frac{V_{in}DT}{L} \frac{1}{T(V_o - V_{in})}$$

$$D_A = \frac{V_{in}D}{L(V_o - V_{in})}$$

Now substituting I_{peak} and D_A in $I_{L,avg}$,

$$I_{L,avg} = \frac{1}{2} \left[D + \frac{V_{in}D}{V_o - V_{in}} \right] \left[\frac{V_{in}D}{L f_s} \right]$$

$$I_{L,avg} = \frac{V_{in}D^2}{2Lf_s} \left[1 + \frac{V_{in}}{V_o - V_{in}} \right]$$

$$I_{L,avg} = \frac{V_{in}D^2}{2Lf_s} \left[\frac{V_o - V_{in} + V_{in}}{V_o - V_{in}} \right]$$

$$I_{L,avg} = \frac{V_{in}D^2}{2Lf_s} \left[\frac{V_o}{V_o - V_{in}} \right]$$

Now substituting $I_{L,avg}$ in power balance,

$$\frac{V_{in}^2 D^2}{2Lf_s} \left[\frac{V_o}{V_{in} - V_o} \right] = \frac{V_o^2}{R}$$

$$\frac{V_{in}^2 D^2 R}{2Lf_s} = V_o(V_{in} - V_o)$$

$$-\frac{V_{in}^2 D^2 R}{2Lf_s} = V_o V_{in} - V_o^2$$

$$-V_o V_{in} + V_o^2 - \frac{V_{in}^2 D^2 R}{2Lf_s} = 0$$

$$V_o = \frac{V_{in} + \sqrt{V_{in}^2 + \frac{2V_{in}^2 D^2 R}{Lf_s}}}{2}$$

$$V_o = \frac{V_{in}}{2} + \frac{V_{in}}{2} \sqrt{1 + \frac{2D^2 R}{Lf_s}}$$

$\therefore$ we will only consider since $V_o > V_{in}$

$$\langle V_o \rangle = \frac{V_{in}}{2} + \frac{V_{in}}{2} \sqrt{1 + \frac{2D^2 R}{Lf_s}}$$

DCM Inductor Current Ripple

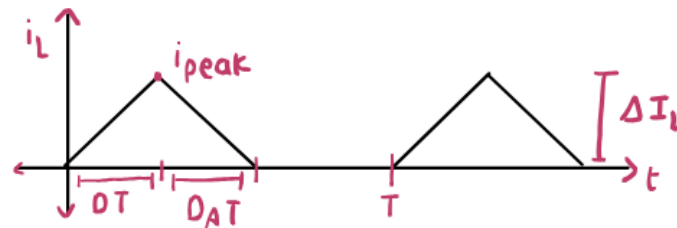


Figure 7: Inductor current ripple dcm

From waveform in DCM,

$$I_{L,\min} = 0$$

During switch ON,

$$V_L = V_{in} \quad V_L = L \frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_{in}}{L}$$

$$I_{peak} = \frac{V_{in}}{L} T_{ON}$$

$$T_{ON} = DT_s = \frac{D}{f_s}$$

$$I_{peak} = \frac{V_{in} D}{L f_s}$$

Since it rises from zero to I_{peak}

Since in DCM,

$$I_{L,\min} = 0$$

$$\Delta I_L = I_{L,\max} - I_{L,\min}$$

$$\Delta I_L = I_{L,\max}$$

where

$$I_{L,\max} = I_{peak}$$

$$\Delta I_L = I_{peak}$$

$$\Delta I_L = \frac{V_{in} D}{L f_s}$$

$$\Delta I_L = \frac{V_{in} D}{L f_s} \begin{cases} \text{CCM}, & I_{L,\min} > 0 \\ \text{DCM}, & I_{L,\min} = 0 \end{cases}$$

Capacitor Voltage Ripple

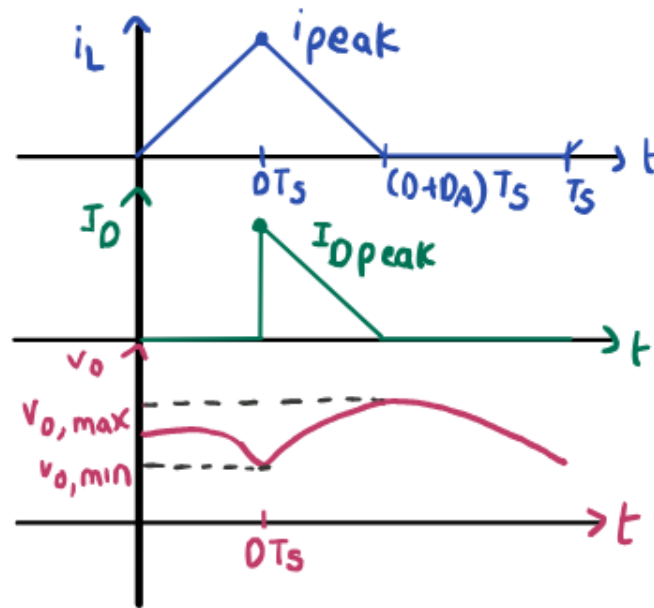


Figure 8: Dcm capacitor voltage ripple

$$\Delta V_C = \frac{I_o D}{RC f_s}$$

0.5 MATLAB Simulations and Calculations

0.5.1 CCM Calculations

Input:

$$V_{in} = 24 V$$

$$D = 0.5$$

$$f_s = 50 kHz$$

$$L = 100 \mu H$$

$$C = 100 \mu F$$

$$R = \text{variable}$$

Critical resistance calculation:

$$R_{crit} = \frac{2Lf_s(1-D)^2}{D}$$

$$= 2(100 \times 10^{-6})(50 \times 10^3) \frac{(0.5)^2}{0.5}$$

$$R_{crit} = 5 \Omega$$

Therefore,

$$\text{CCM: } R = 2 \Omega$$

$$\text{Boundary: } R = 5 \Omega$$

$$\text{DCM: } R = 20 \Omega$$

Case I: CCM

$$R = 2 \Omega$$

$$\langle V_o \rangle = \frac{V_{in}}{1 - D}$$

$$= \frac{24}{1 - 0.5}$$

$$\langle V_o \rangle = 48 V$$

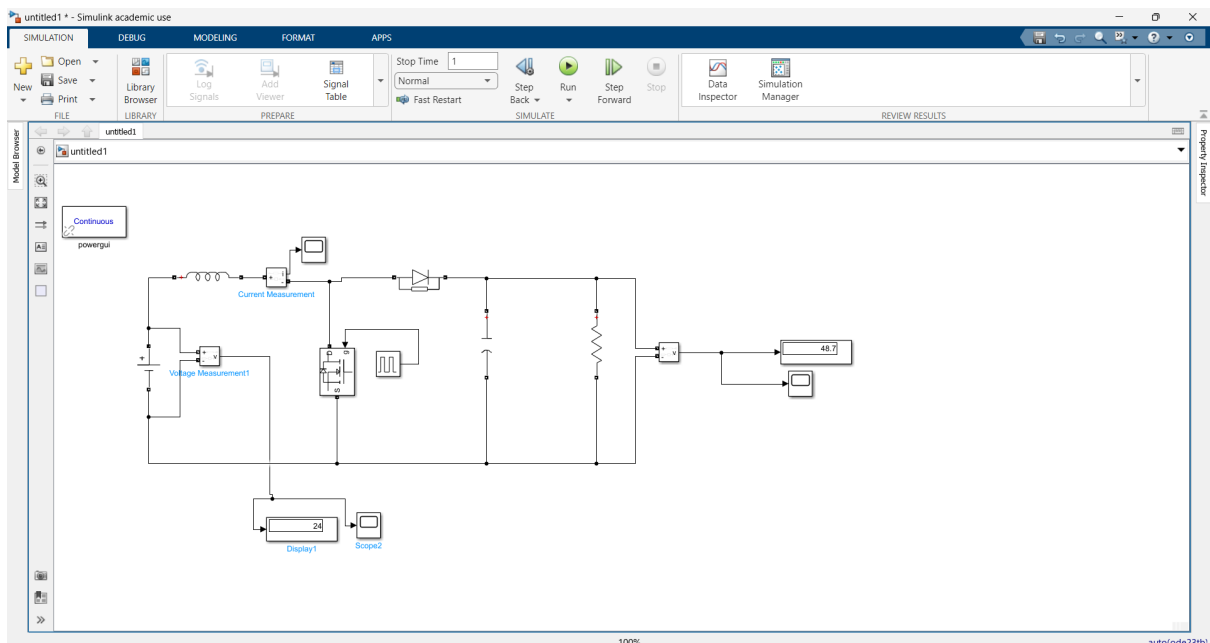


Figure 9: CCM Average Output Voltage

$$\Delta I_L = \frac{V_{in} D}{L f_s}$$

$$= \frac{24(0.5)}{(100 \times 10^{-6})(50 \times 10^3)}$$

$$\Delta I_L = 2.4 \text{ A}$$

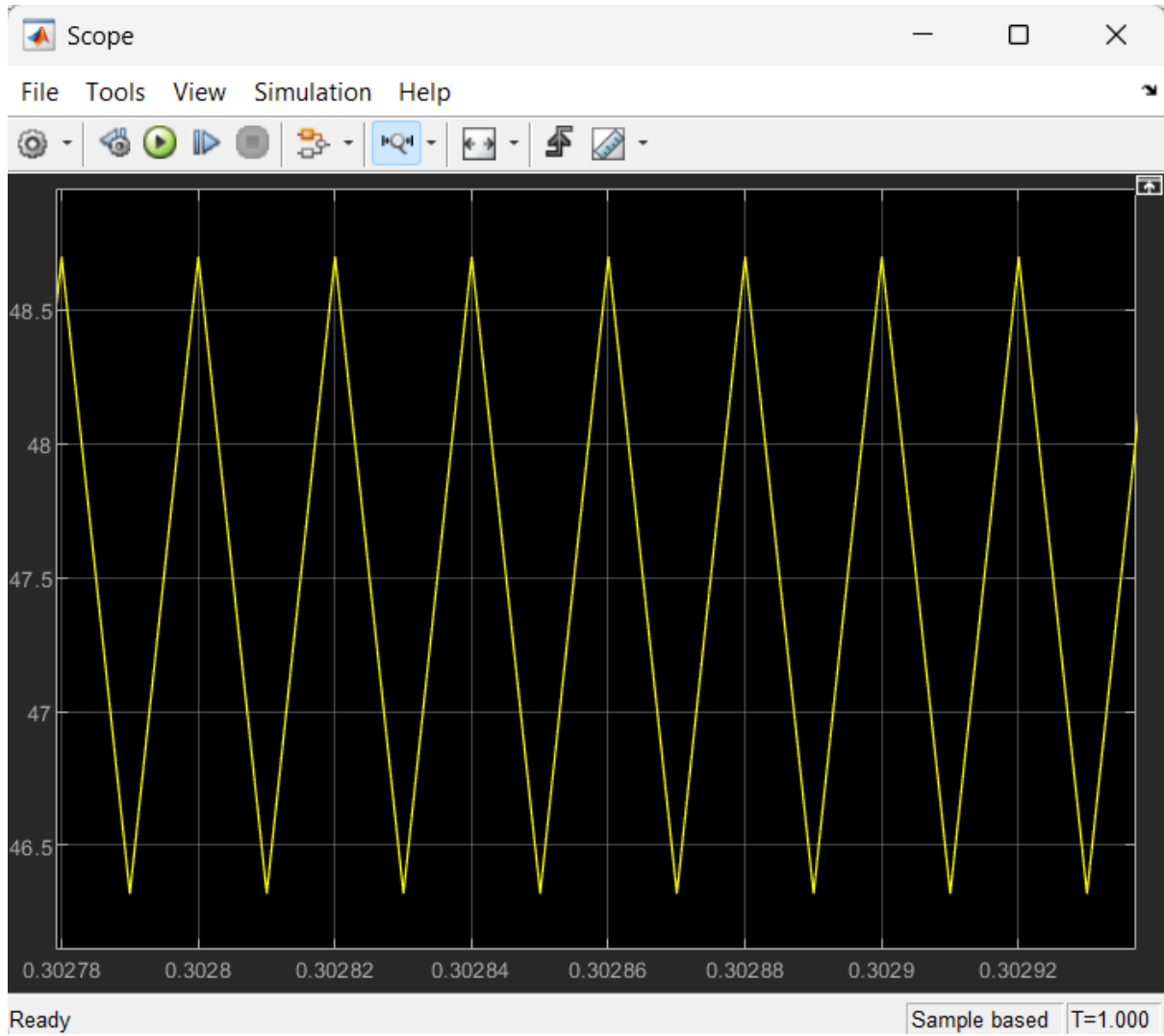


Figure 10: CCM Inductor Current Ripple

$$\Delta V_C = \frac{I_o D}{C f_s}$$

$$I_o = \frac{V_o}{R} = \frac{48}{2} = 24 \text{ A}$$

$$\Delta V_C = \frac{24(0.5)}{(100 \times 10^{-6})(50 \times 10^3)}$$

$$\Delta V_C = 2.4 \text{ V}$$

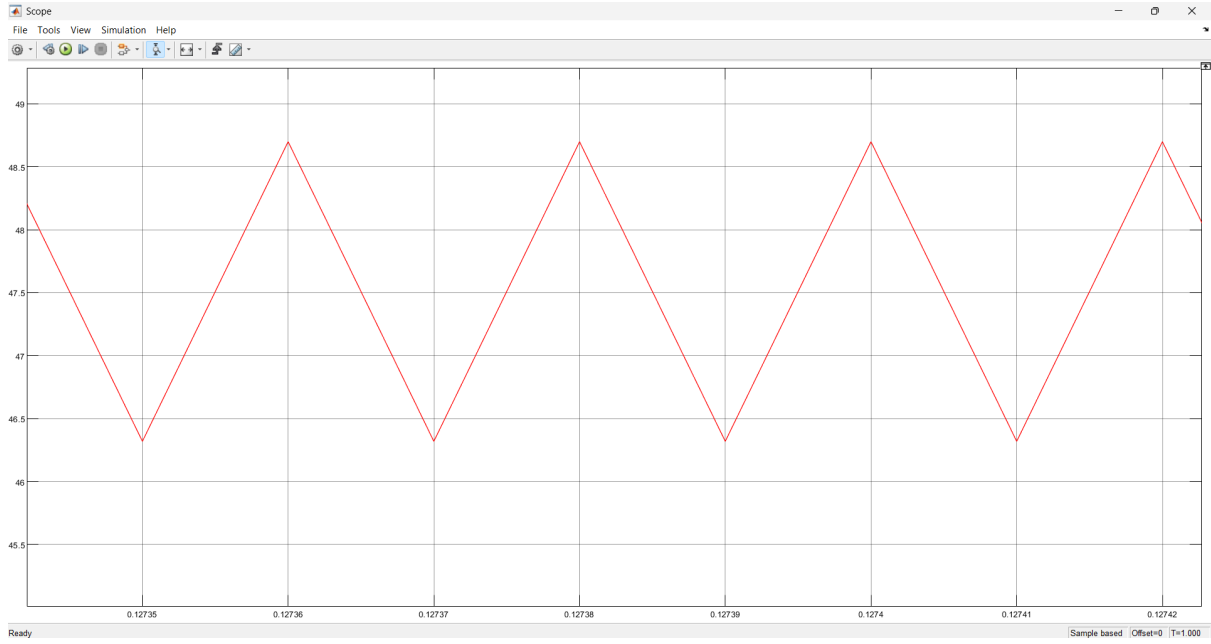


Figure 11: CCM Capacitor Voltage Ripple

0.5.2 DCM Calculations

Case II: DCM

$$R = 20 \, \Omega$$

$$\begin{aligned} \langle V_o \rangle &= \frac{V_{in}}{2} + \frac{V_{in}}{2} \sqrt{1 + \frac{2D^2 R}{L f_s}} \\ &= \frac{24}{2} \left[1 + \sqrt{1 + \frac{2(0.5)^2 (20)}{(100 \times 10^{-6}) (50 \times 10^3)}} \right] \\ &= 12 \left[1 + \sqrt{3} \right] \end{aligned}$$

$$\langle V_o \rangle = 32.78 \, V$$

$$\langle V_o \rangle_{CCM} > \langle V_o \rangle_{DCM}$$

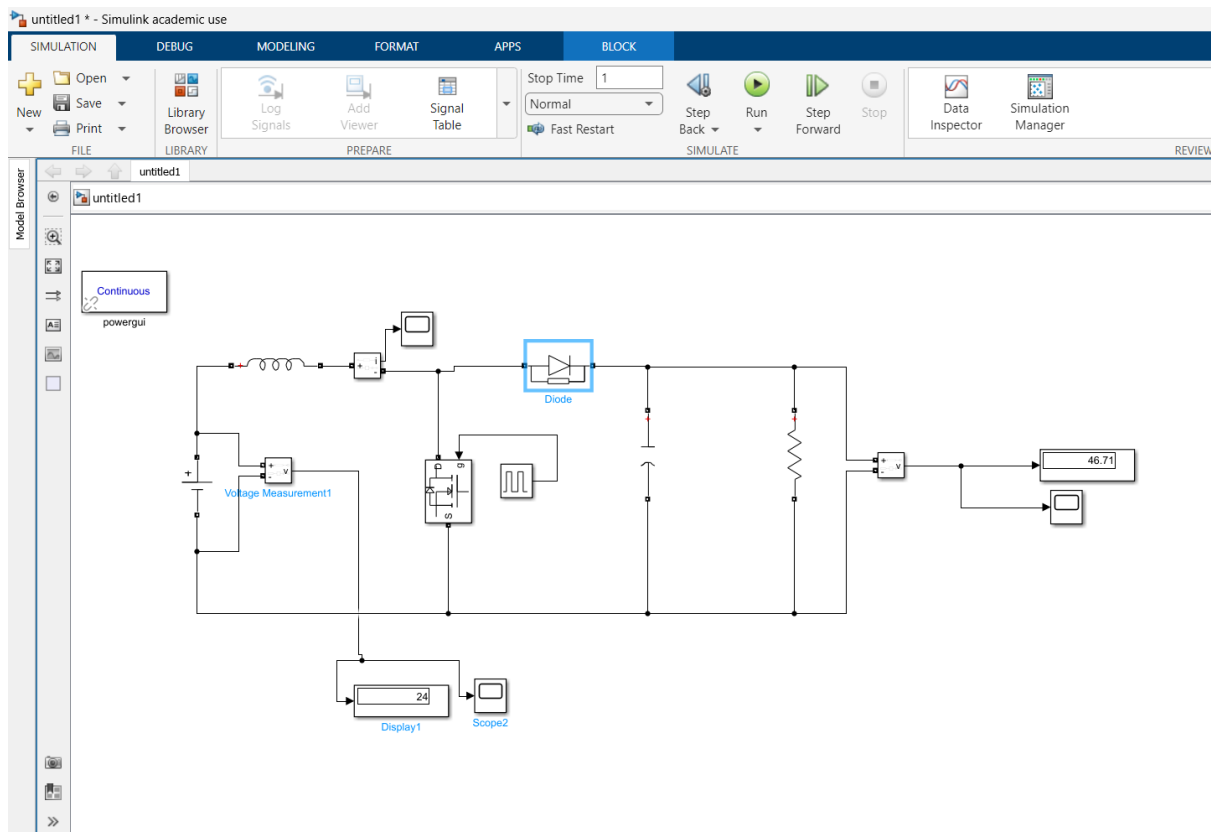


Figure 12: DCM Average Output Voltage

$$\Delta I_L = \frac{V_{in} D}{L f_s}$$

$$= \frac{24(0.5)}{(100 \times 10^{-6})(50 \times 10^3)}$$

$$\Delta I_L = 2.4 \text{ A}$$

$$I_{L,\min} = 0$$

$$I_{L,\max} = 2.4 \text{ A}$$

$$\Delta V_C = \frac{I_o D}{C f_s}$$

$$I_o = \frac{V_o}{R} = \frac{32.78}{20} = 1.639 \text{ A}$$

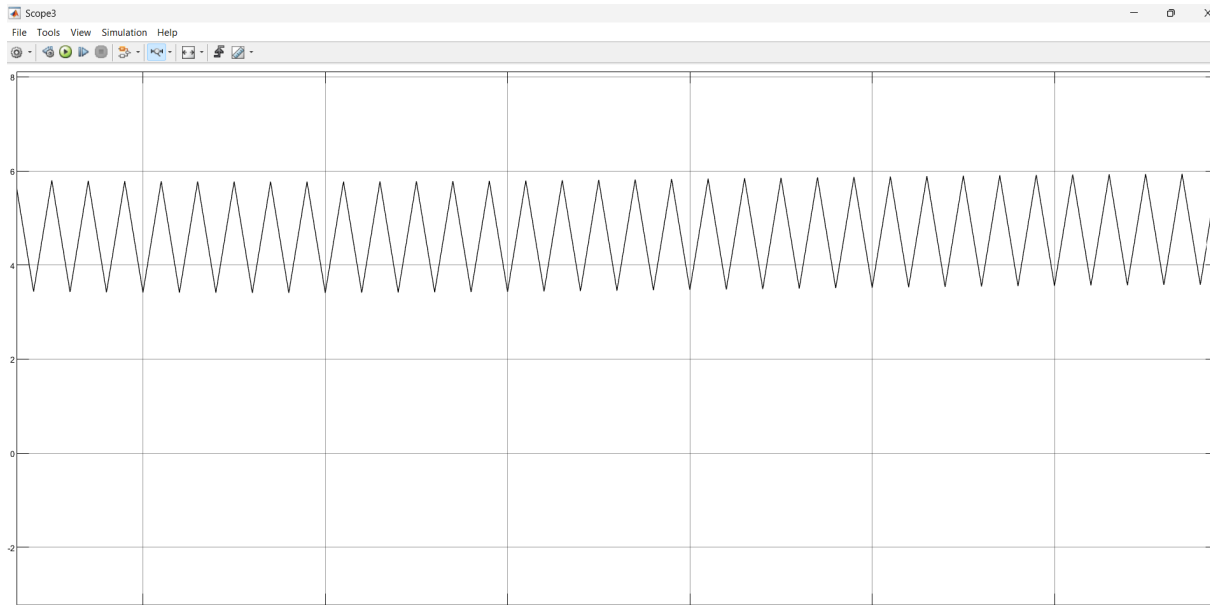


Figure 13: DCM Inductor Current Ripple

$$\Delta V_C = \frac{1.639(0.5)}{(100 \times 10^{-6})(50 \times 10^3)}$$

$$\Delta V_C = 0.164 \text{ V}$$

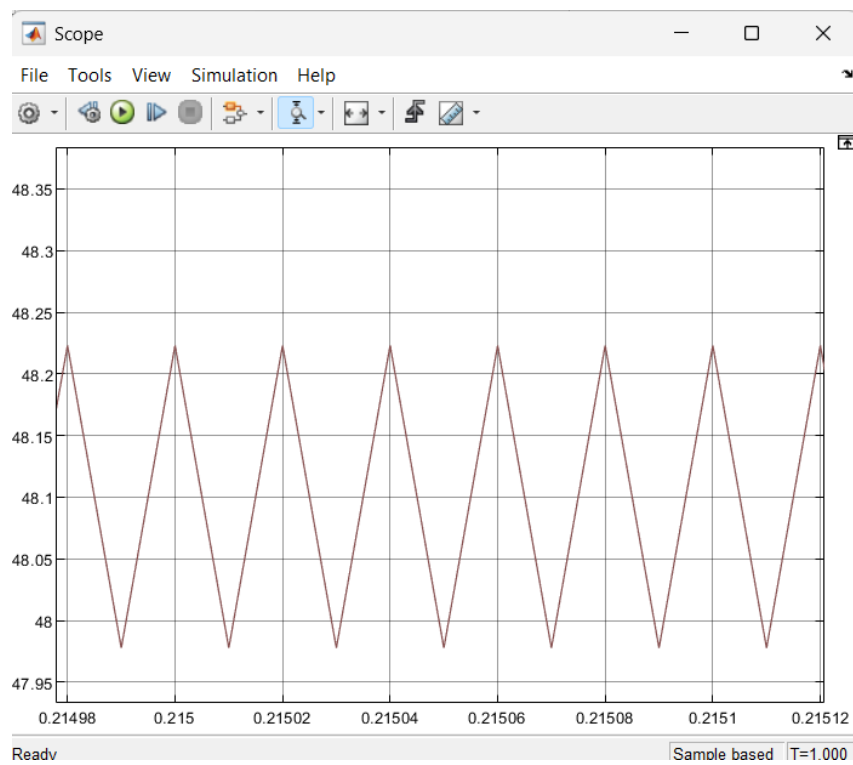


Figure 14: DCM Capacitor voltage ripple

0.5.3 Boundary Case

$$R = R_{crit} = 5 \Omega$$

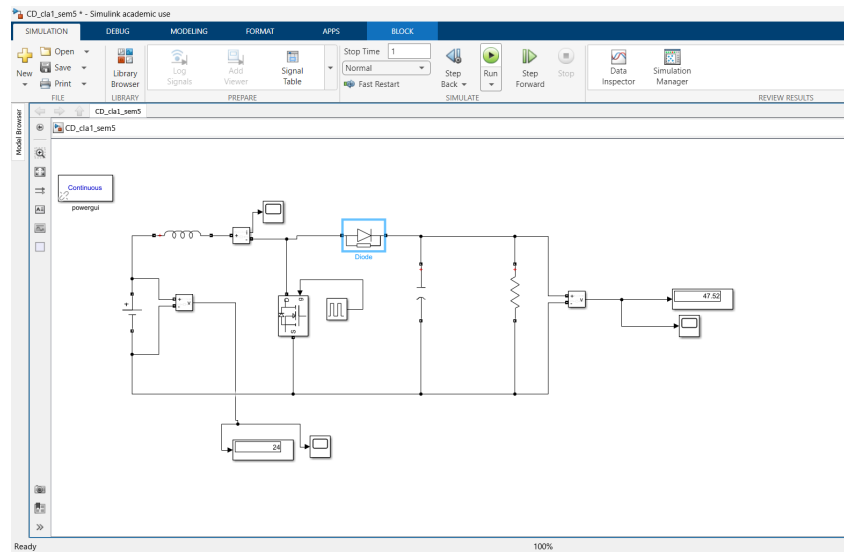


Figure 15: Boundary condition output voltage

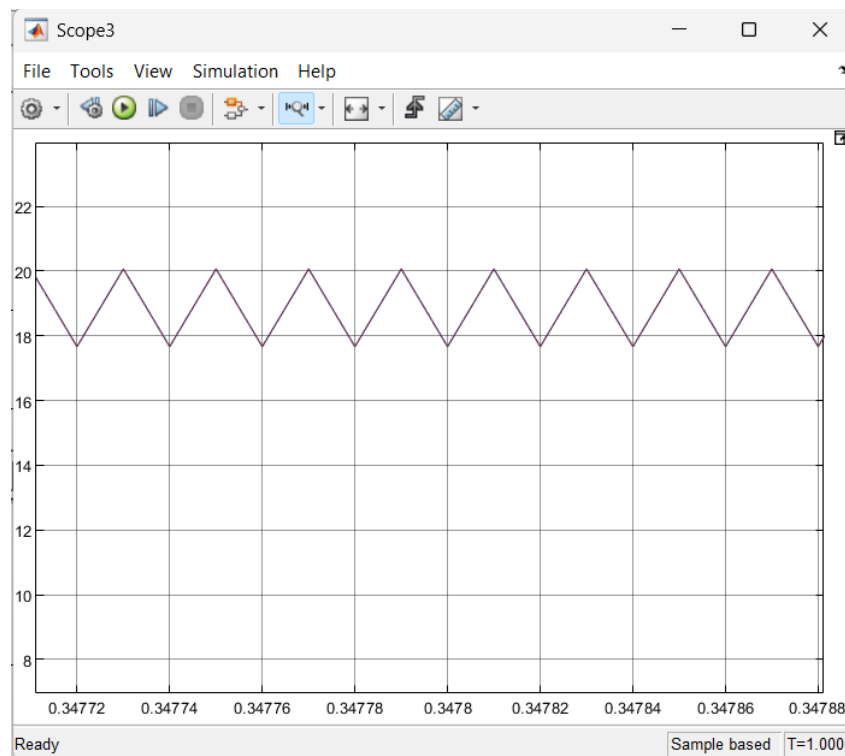


Figure 16: Boundary condition Inductor current ripple

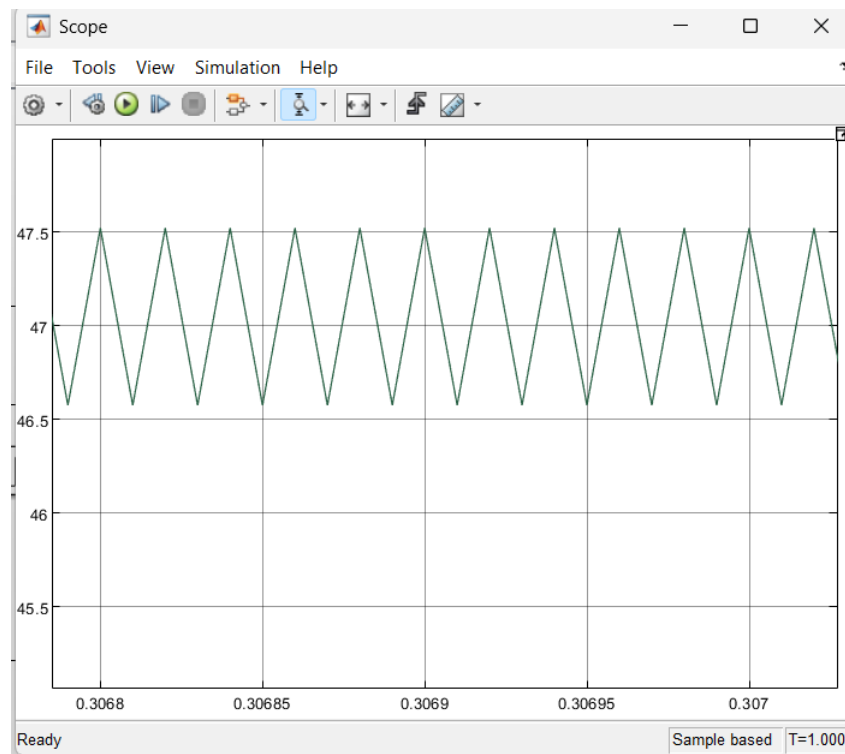


Figure 17: Boundary condition capacitor voltage ripple